

IN-CLASS ACTIVITY

Build a Mini E-Commerce Website for Business Data Mining

Business Data and Text Mining | Duration: 60 Minutes | Individual / Pair Activity

BUSINESS SCENARIO

You are launching a fictional online store. Your webpage will contain product data that a Business Analytics team could later collect, scrape, and analyze. Today you build the data source; in a later class, the same structure can be mined using Python and BeautifulSoup.

Learning Objectives

- Use the basic HTML document structure: `<html>`, `<head>`, `<title>`, `<style>`, and `<body>`.
- Use internal CSS to style a business webpage.
- Represent business data consistently using HTML classes.
- Recognize how websites can act as sources of structured and semi-structured business data.
- Connect HTML structure with future web scraping and business data mining.

Your Mission

Create one file named **my_store.html**. Choose a fictional business such as a tech store, clothing brand, bookstore, gaming store, cosmetics store, or food business. Make the page attractive, consistent, and easy for a future data-mining program to understand.

60-Minute Challenge

Time	Stage	Task
0–10 min	1. Launch Your Store	Create the HTML structure, store name, heading, tagline, and short business description.
10–30 min	2. Build the Dataset	Add at least 6 products. Every product must use the same set of business-data fields and CSS classes.
30–45 min	3. Design the Store	Use the <code><style></code> tag to make the page look like a real online store.
45–55 min	4. Business Analysis	Answer the six business questions below by examining your webpage.
55–60 min	5. Showcase	Open selected pages, compare designs, and discuss how Python could automatically collect the data.

Required Product Data

Each of your 6+ products must contain all of the following:

Field	Example
Product Name	Wireless Headphones
Category	Electronics
Price	Rs. 7,999
Rating	4.6 / 5

Availability	In Stock
Number of Reviews	128
Description	Noise-cancelling wireless headphones

HTML Data Rules

Use consistent classes so that the information could later be extracted automatically. Your products should use:

- `class="product"`
- `class="name"`
- `class="category"`
- `class="price"`
- `class="rating"`
- `class="availability"`
- `class="reviews"`
- `class="description"`

CSS Design Requirements

Inside the `<style>` tag, style at least: **body**, **h1**, **.product**, **.price**, **.rating**, and **.availability**. Use several of the following properties:

- `background-color` and `color`
- `font-family` and `font-size`
- `border`
- `padding` and `margin`
- `text-align`
- `width`

Design Challenge: Make the products look like cards and make In Stock and Out of Stock visually different.

Business Data Challenge — Answer in HTML Comments

1. Which product is the most expensive?
2. Which product has the highest rating?
3. Which category has the most products?
4. How many products are out of stock?
5. Which product has the most reviews?
6. If you were the store manager, which product would you promote and why?

Bonus Challenges — Unlock Extra Points

★ Level 1 — Sale Badge

Add a SALE or BEST SELLER label to one product and style it using a new CSS class.

★★ Level 2 — Hover Effect

Make a product card or button change appearance when the mouse moves over it using :hover.

★★★ Level 3 — Customer Review

Add one short customer review to each product. Keep the same class name for every review so the text could later be mined.

★★★★ Level 4 — Data Quality Trap

Intentionally create ONE bad data value, such as a missing rating or inconsistent price format. Be ready to explain why this could cause problems during data analysis.

The Final Question

What if the website contained 10,000 products instead of 6?

How could Python automatically find every product, price, rating, review, and availability value?

Class Discussion

Look at this idea: If every price uses class="price", a program does not need to understand the visual design of the webpage. It can search the HTML structure for the data it needs.

Connection to the course: HTML → Web Data → Web Scraping → DataFrame → Analysis → Text Mining → Business Decision

Submission

- Submit the single my_store.html file.
- Minimum 6 products with all required fields.
- Include your answers to the six business questions as HTML comments at the bottom of the file.
- The webpage must open correctly in a browser.

Assessment — 20 Marks

Criterion	Marks
Correct HTML structure	4
Correct and consistent business data	4
Use of internal CSS	4
Design and readability	3
Business-analysis answers	3
Creativity / bonus challenge	2
TOTAL	20

Build it today. Mine it later.